Matrix functions and stability

Multiplication overhead of cubic sign compositions

Difficulty: challenging **Importance:** interesting to the community

Status: open; explicit 2026 problem

Context and notation

For $m \in \mathbb{N}_0$ and $0 < \delta < 1$, put

$$I_\delta = [-1, -\delta] \cup [\delta, 1].$$

Let $\mathcal{P}_m$ consist of real polynomials computed from $1, x$ by straight-line programs using at most m nonscalar multiplications; real linear combinations cost nothing. Define

$$E_m(\delta) = \inf_{p \in \mathcal{P}_m} \max_{x \in I_\delta} |p(x) - \operatorname{sign}(x)|.$$

The arithmetic model concerns a single polynomial identity valid for matrices of every size.

Problem statement

Define

$$C_T(\delta) = \inf_{a_t, b_t \in \mathbb{R}} \max_{x \in I_\delta} |(q_T \circ \cdots \circ q_1)(x) - \operatorname{sign}(x)|, \qquad q_t(x) = a_t x + b_t x^3.$$

Determine the asymptotic dependence on m, δ of

$$T_{\min}(m, \delta) = \inf\{T \in \mathbb{N}_0 : C_T(\delta) \le E_m(\delta)\},$$

where the empty composition is x and $\inf \emptyset = +\infty$. Each stage costs at most two matrix products.

References and status evidence

Amsel et al., Simons workshop report, §6.3, Problem 6.5. Rubensson, Jarlebring and Lorentzon, degree-eight recursive expansion, §7, provides a June 2026 follow-up discussion; it does not settle this comparison.

Scope

MF-01 asks for an optimum value over general evaluation programs. This problem measures the price of a particular, reusable composition architecture; the two tasks are related but have different requested outputs.